

WINDOWS SERVER & ACTIVE DIRECTORY MULTI-DOMAIN LAB

Practical Infrastructure & Systems Administration Evidence

Author: Mohammad Arif

Technology Platform: Microsoft Hyper-V

Core Technologies: Windows Server

Active Directory Domain Services

DNS

Group Policy

Domain Authentication

Windows Client Administration

Lab Environment: Root Domain + Child Domain + Replica / Secondary Domain + Separate Tree Domain

Practical Hands-On Project

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1. Project Overview

This project involved designing, deploying and configuring a multi-domain Windows Server environment using Microsoft Hyper-V.

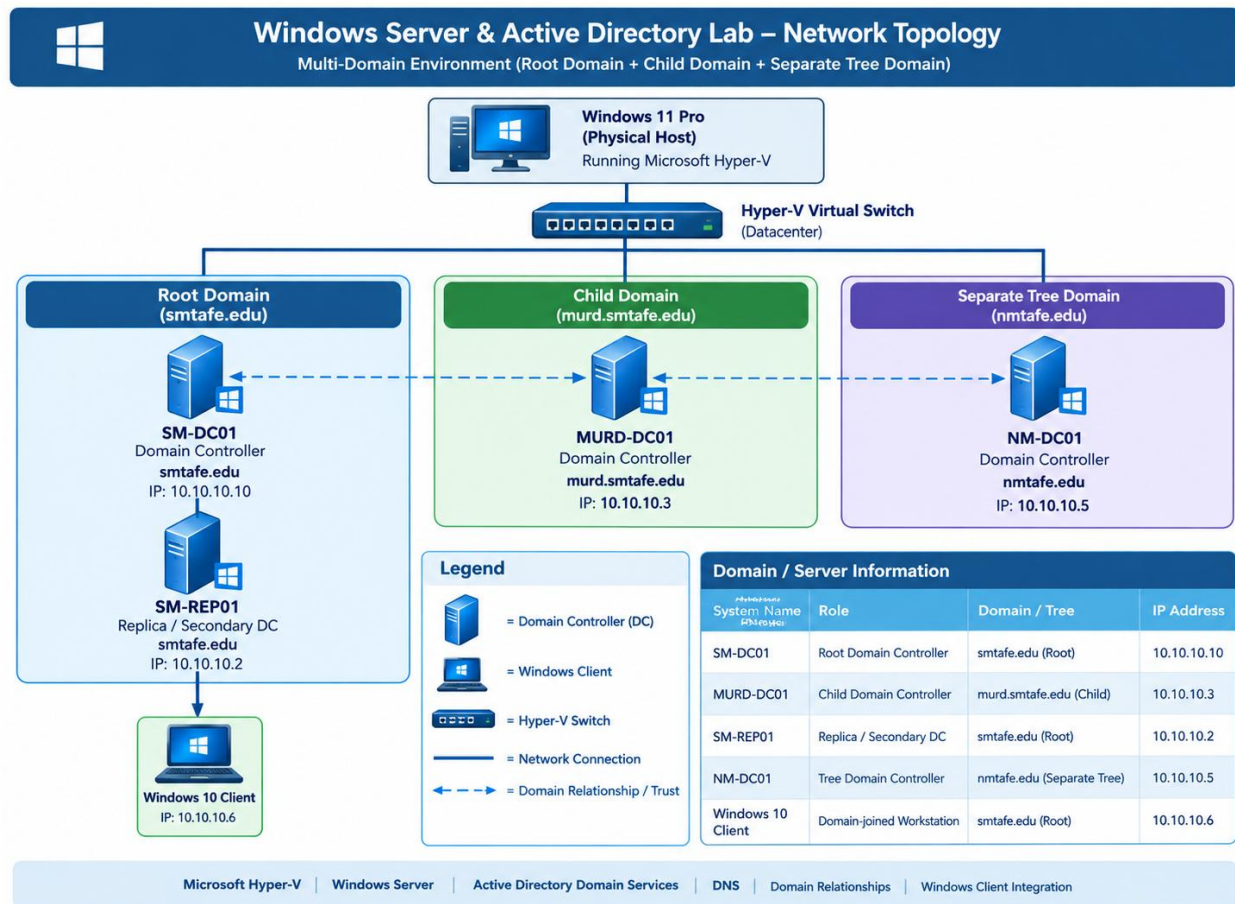
The lab consisted of a root Active Directory domain, a child domain, a separate Active Directory tree/domain, a replica/secondary server and Windows client computers joined to the domain environment. The project provided practical experience with Windows Server administration, Active Directory Domain Services (AD DS), DNS, domain relationships, user and computer administration, domain authentication and Windows client integration. All servers and client machines were created as virtual machines within Microsoft Hyper-V and configured to communicate within the laboratory network.

2. Lab Objectives

The objectives of the lab were to:

- ✓ Deploy Windows Server virtual machines using Hyper-V
- ✓ Configure a Windows Server as the root domain controller ✓ Configure the Active Directory forest ✓ Configure DNS
- ✓ Deploy a child domain
- ✓ Deploy a separate Active Directory tree/domain
- ✓ Configure a replica/secondary domain controller
- ✓ Configure Active Directory organizational structures
- ✓ Create and manage users and groups
- ✓ Configure domain authentication
- ✓ Join Windows client computers to the domain
- ✓ Test domain logon and authentication
- ✓ Manage domain computers and users
- ✓ Verify communication between domain systems
- ✓ Troubleshoot Windows Server and Active Directory services

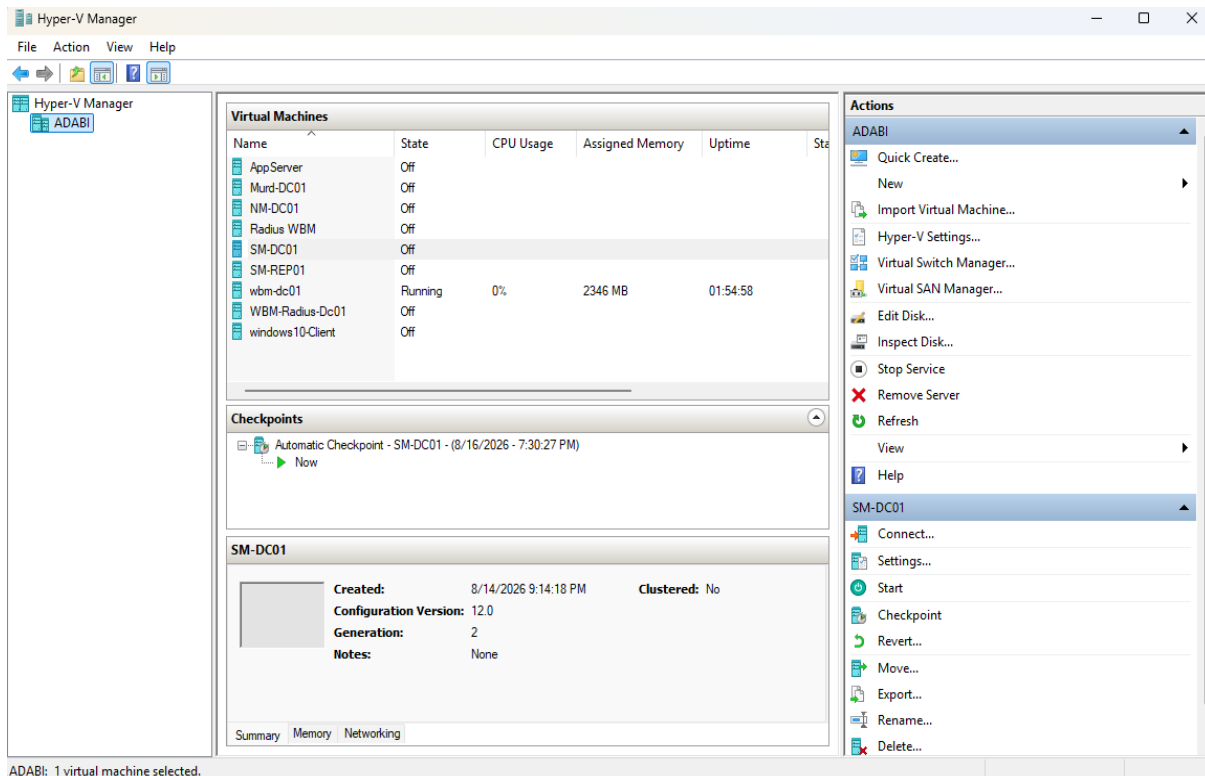
3. Lab Topology



System	Role	Domain
SM-DC01	Root Domain Controller	smtafe.edu
Murd-DC01	Child Domain Controller	murd.smtafe.edu
SM-REP01	Replica/Secondary Server	smtafe.edu
NM-DC01	Tree Domain Controller	nmtafe.edu
Windows Client	Domain-joined workstation	smtafe.edu

4. Hyper-V Virtualization Environment

Figure 1 Hyper-V Manager showing the virtual machines used in the Active Directory laboratory.



The laboratory infrastructure was deployed as multiple virtual machines using Microsoft Hyper-V. Each virtual machine was configured according to its assigned Windows Server or client role.

5. Root Active Directory Domain

Figure 2 Server Manager

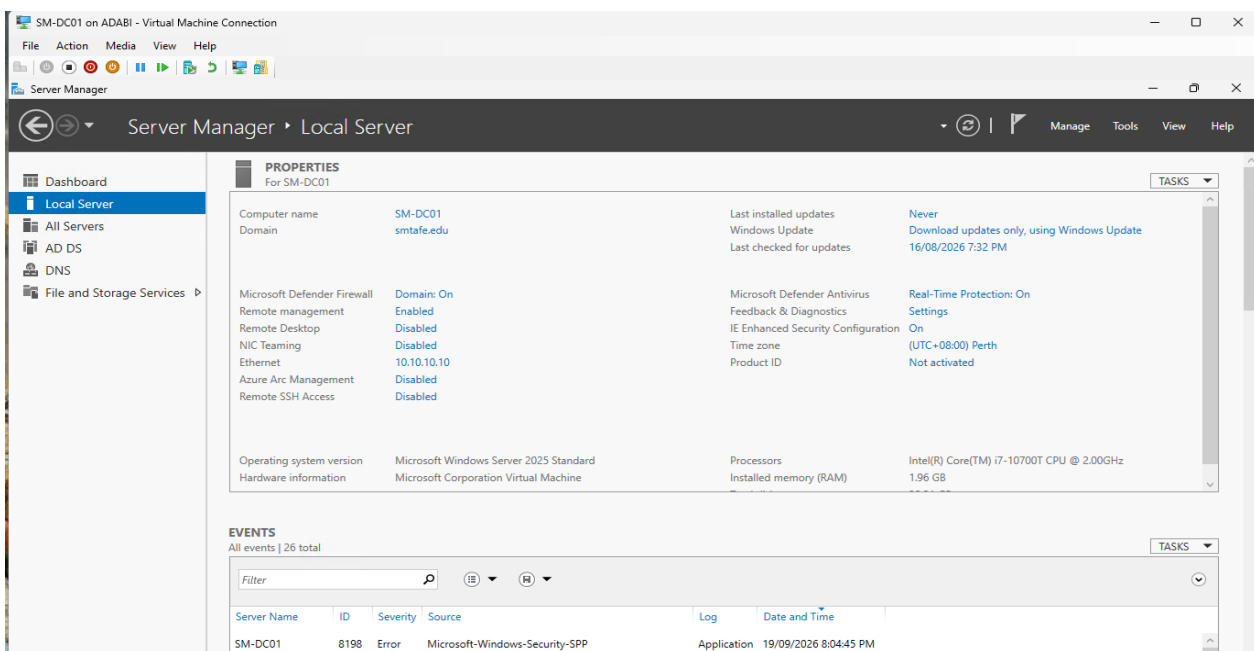


Figure 3-Active Directory Users and Computers

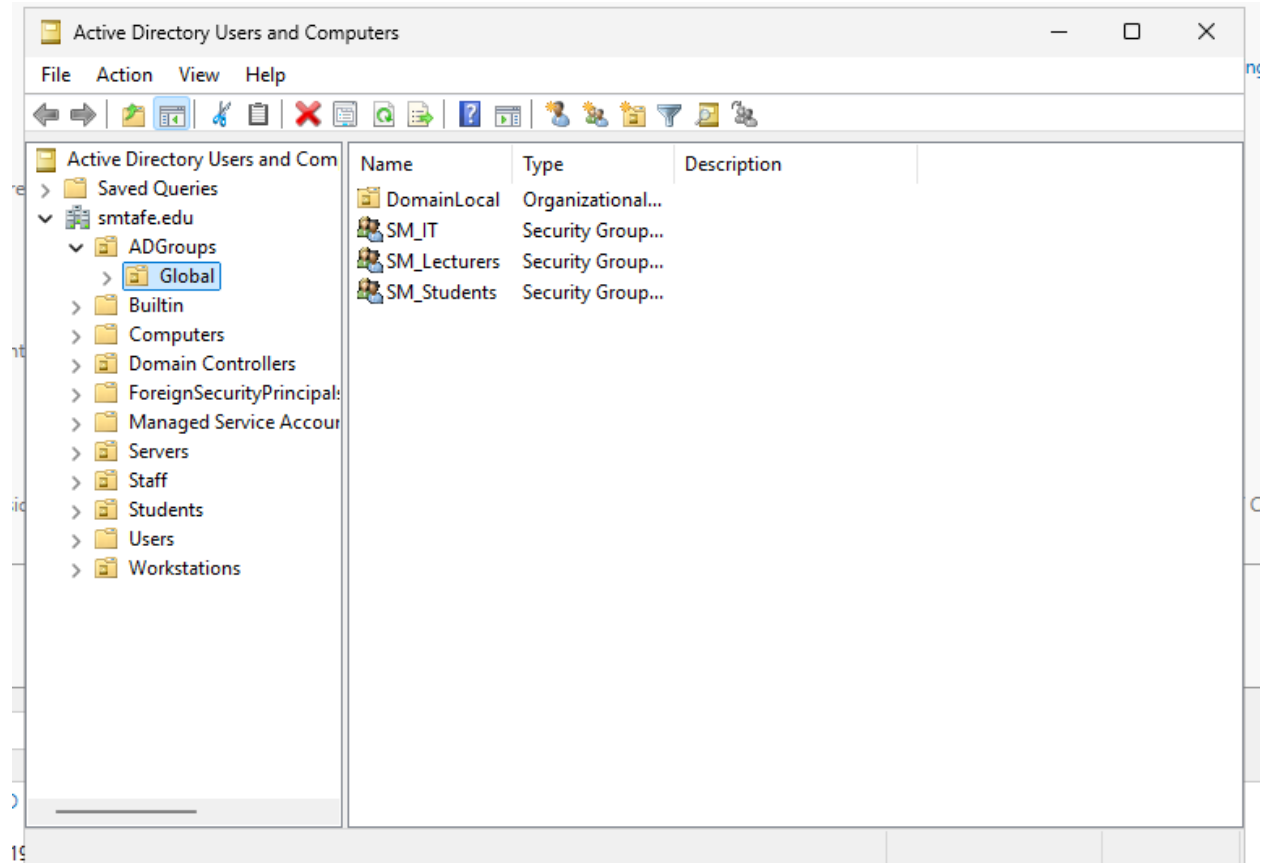
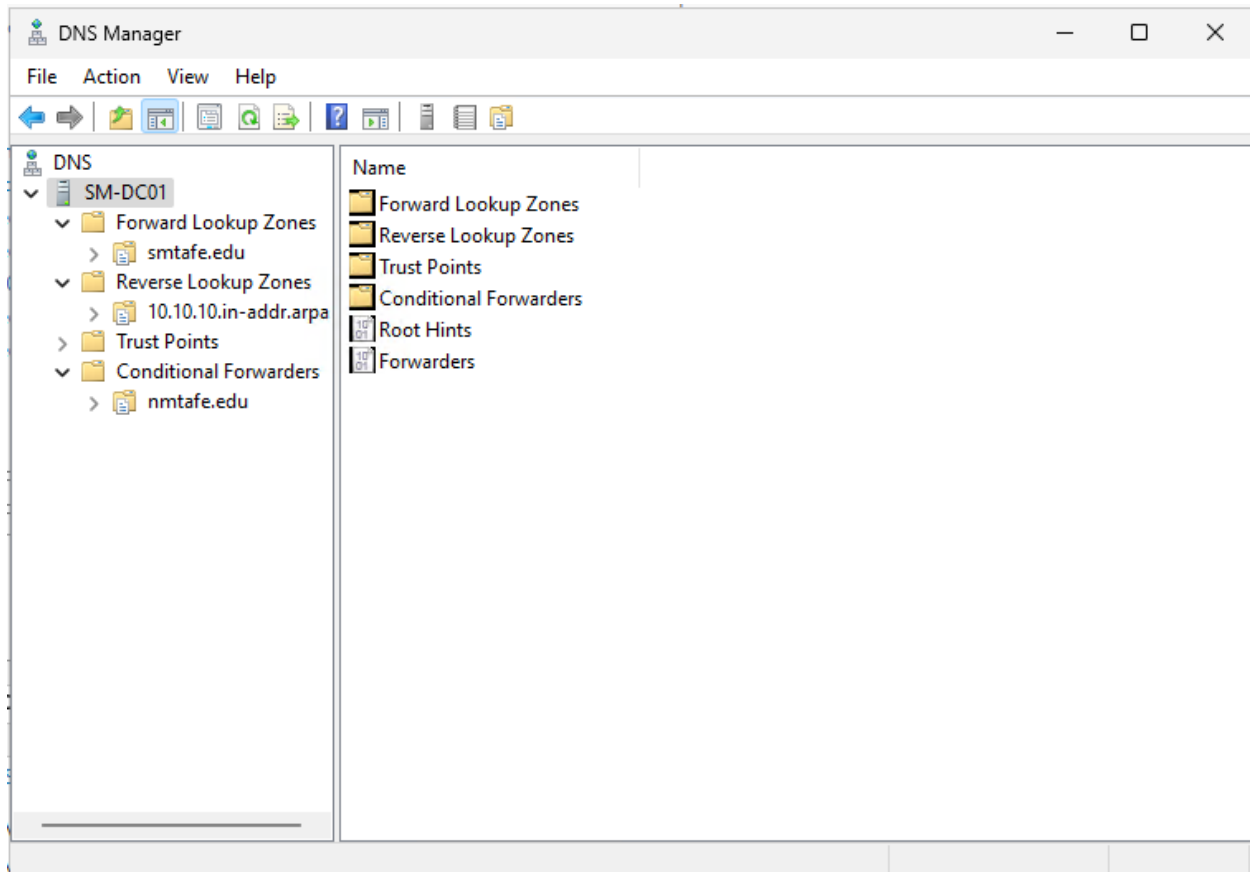


Figure 4-DNS Manager



Evidence: The screenshots demonstrate the configuration of the root Active Directory domain, including Active Directory Domain Services and DNS.

6. Child Domain Configuration

Figure 5-Server Manager

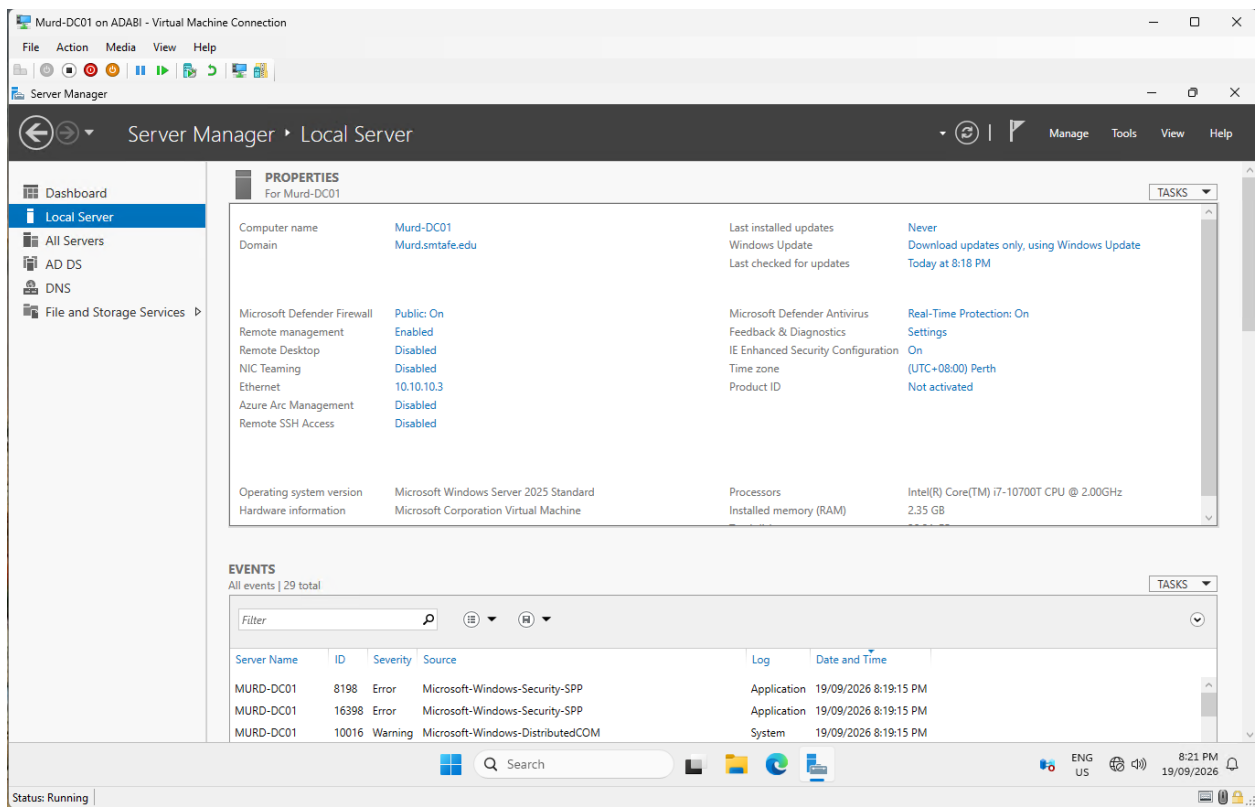


Figure 6-AD Users and Computers

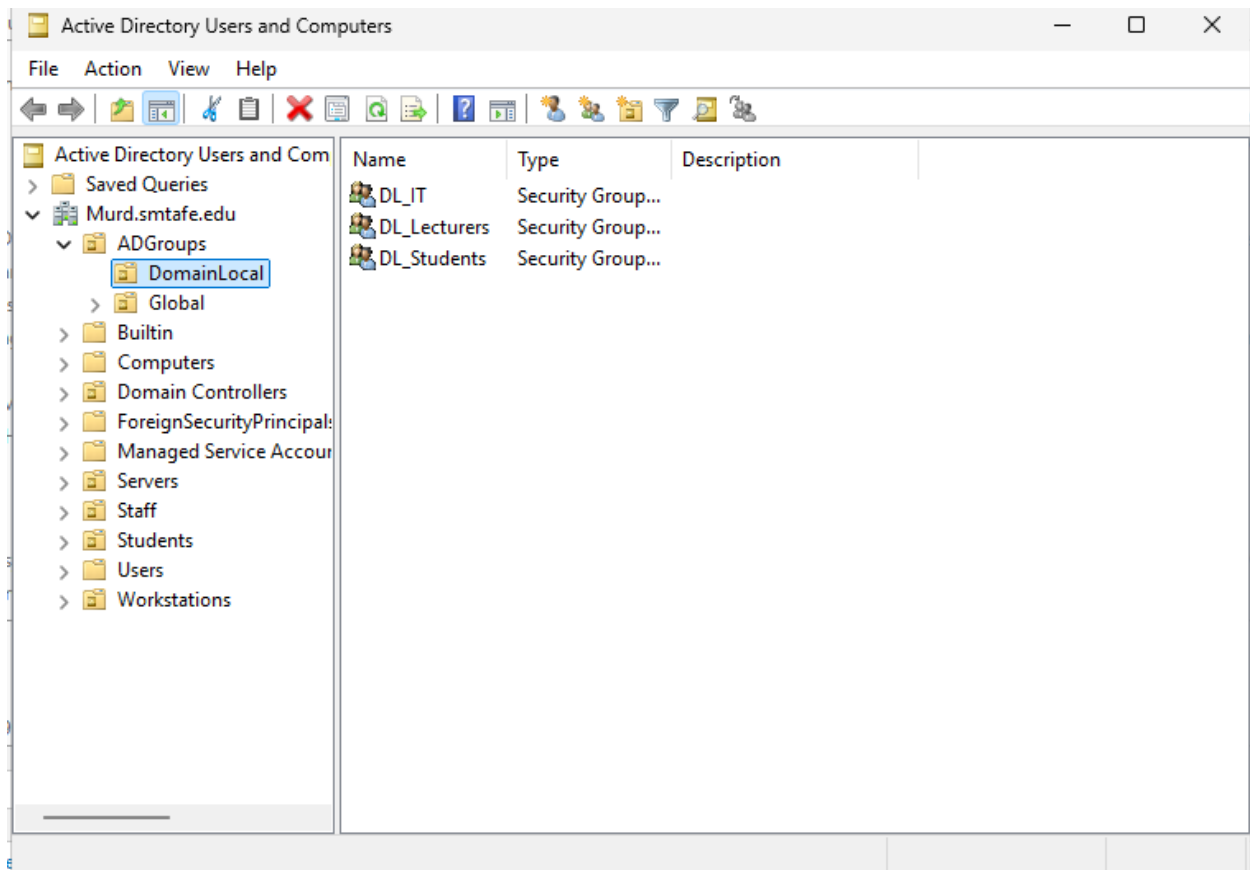


Figure 7-DNS Manager

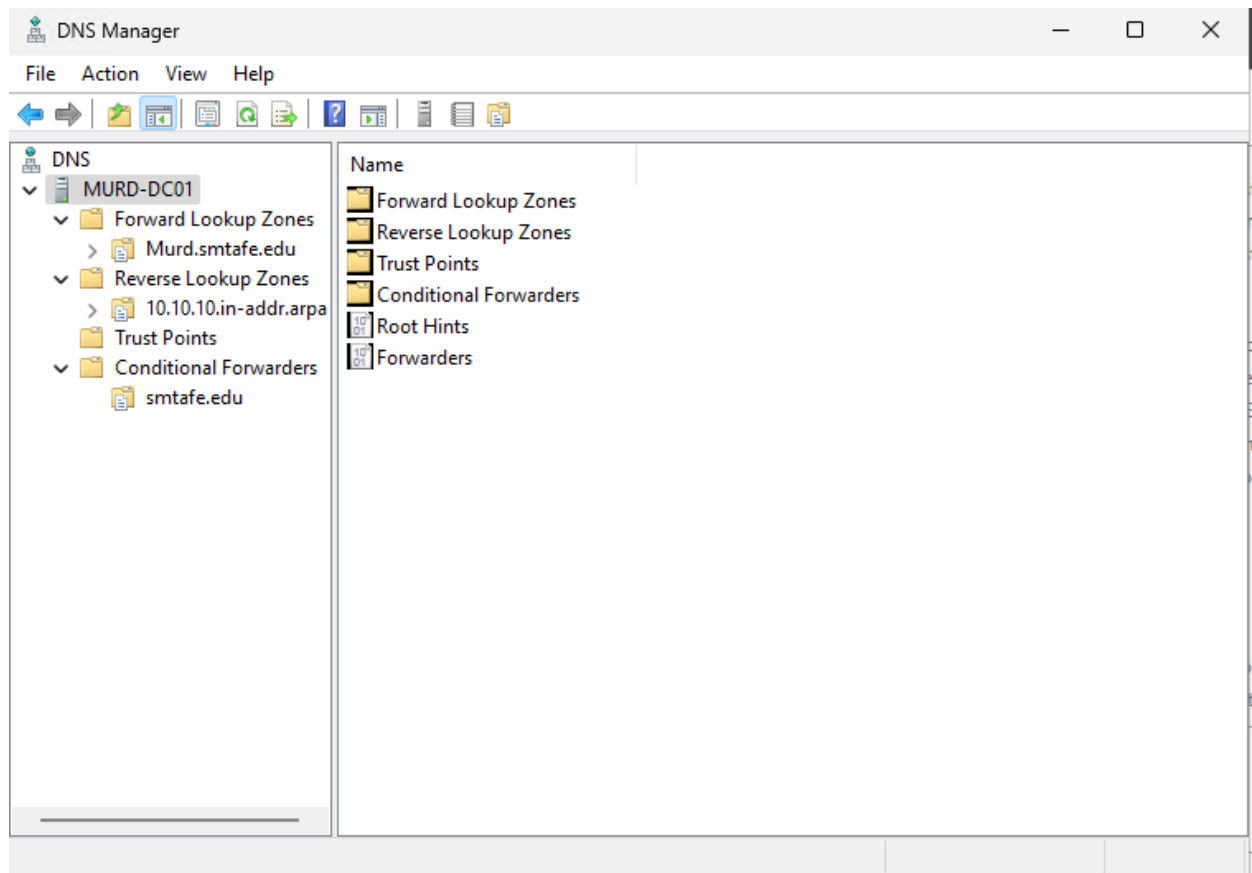


Figure 8-System/domain information

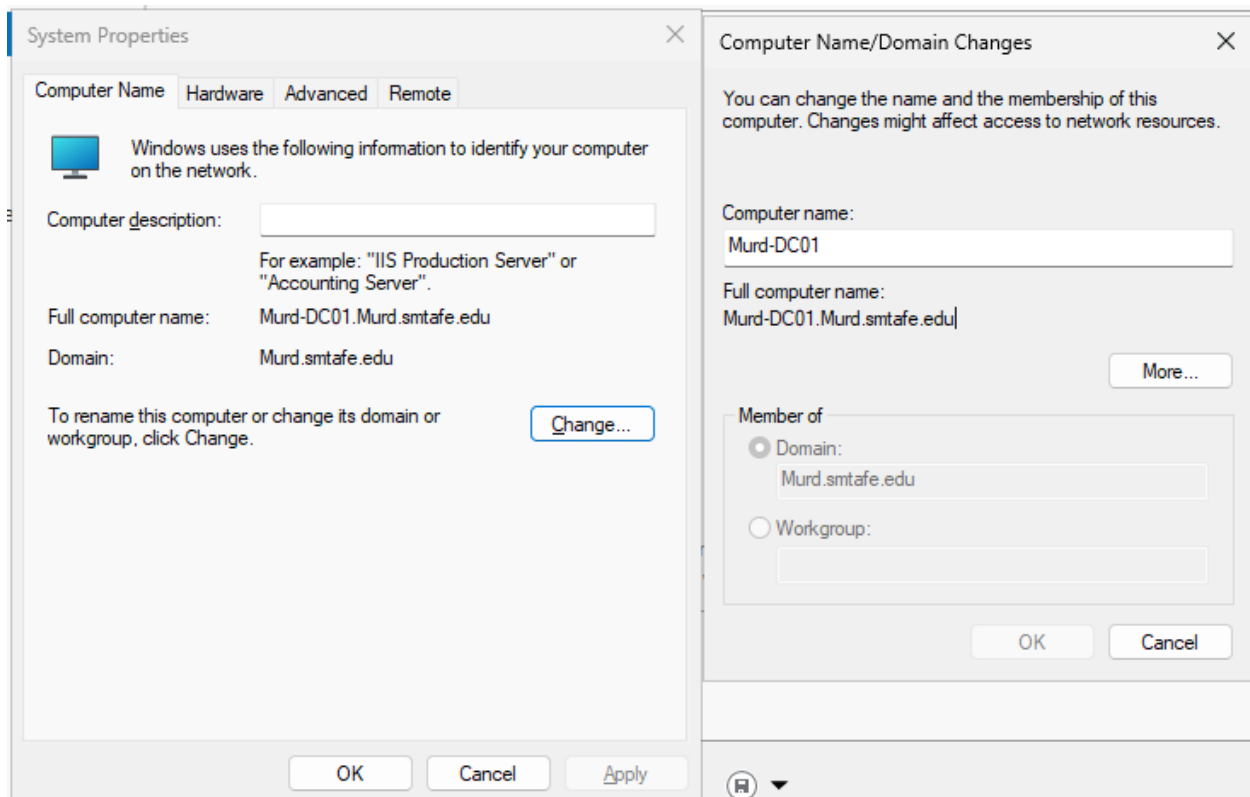
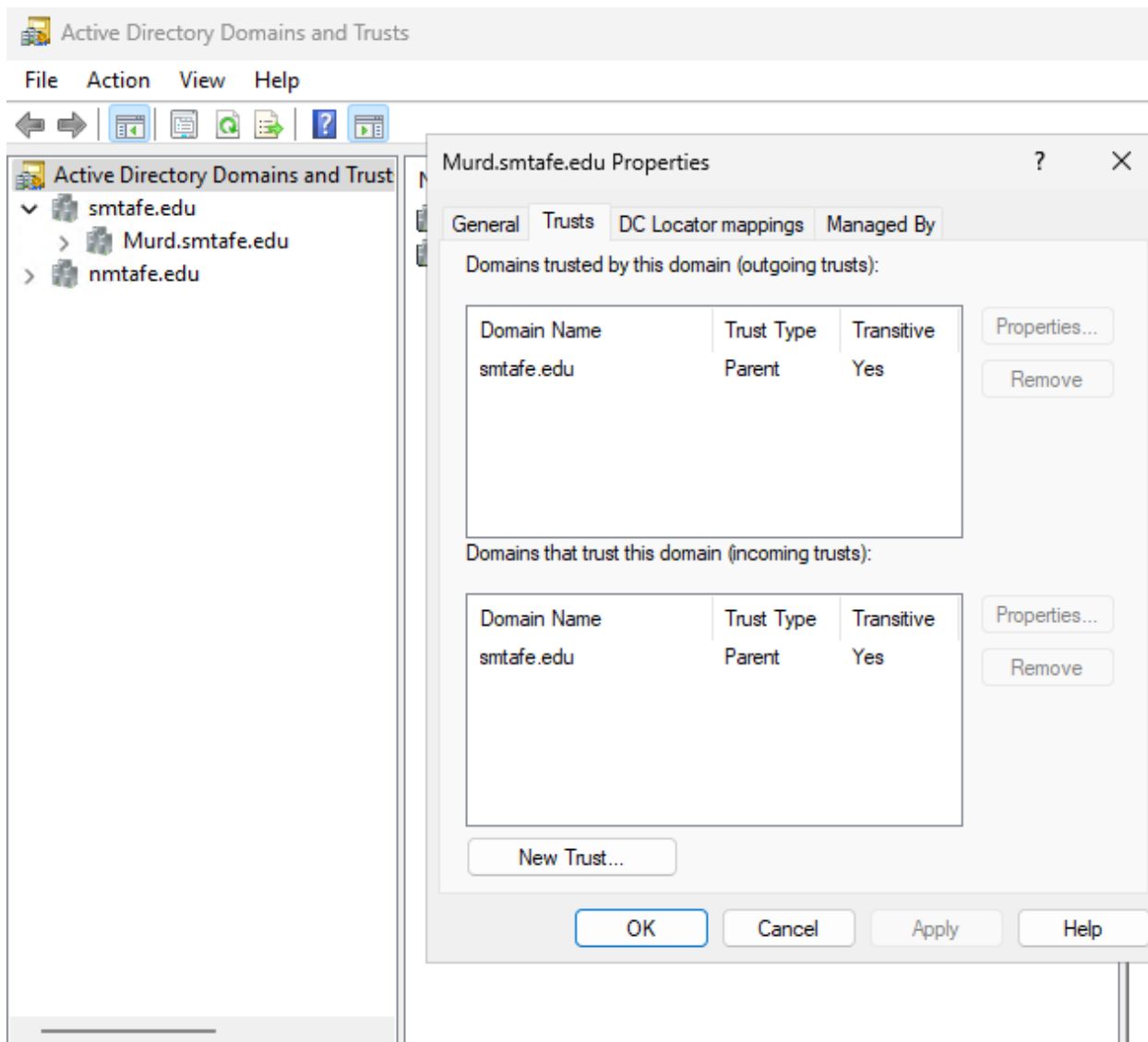


Figure 9-Domain relationship



```

Administrator: Command Pro x + v
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nlttest /domain_trusts
List of domain trusts:
 0: SMTAFE smtafe.edu (NT 5) (Forest Tree Root) (Direct Outbound) (Direct Inbound) ( Attr: withinforest )
 1: MURD Murd.smtafe.edu (NT 5) (Forest: 0) (Primary Domain) (Native)
 2: NMTAFE nmtafe.edu (NT 5) (Forest Tree Root)
The command completed successfully

C:\Users\Administrator>

```

The child domain was deployed under the root domain, creating the hierarchical Active Directory structure **murd.smtafe.edu** within the **smtafe.edu** forest.

7. Replica / Secondary Domain Controller

SM-REP01

Figure 10-Server Manager

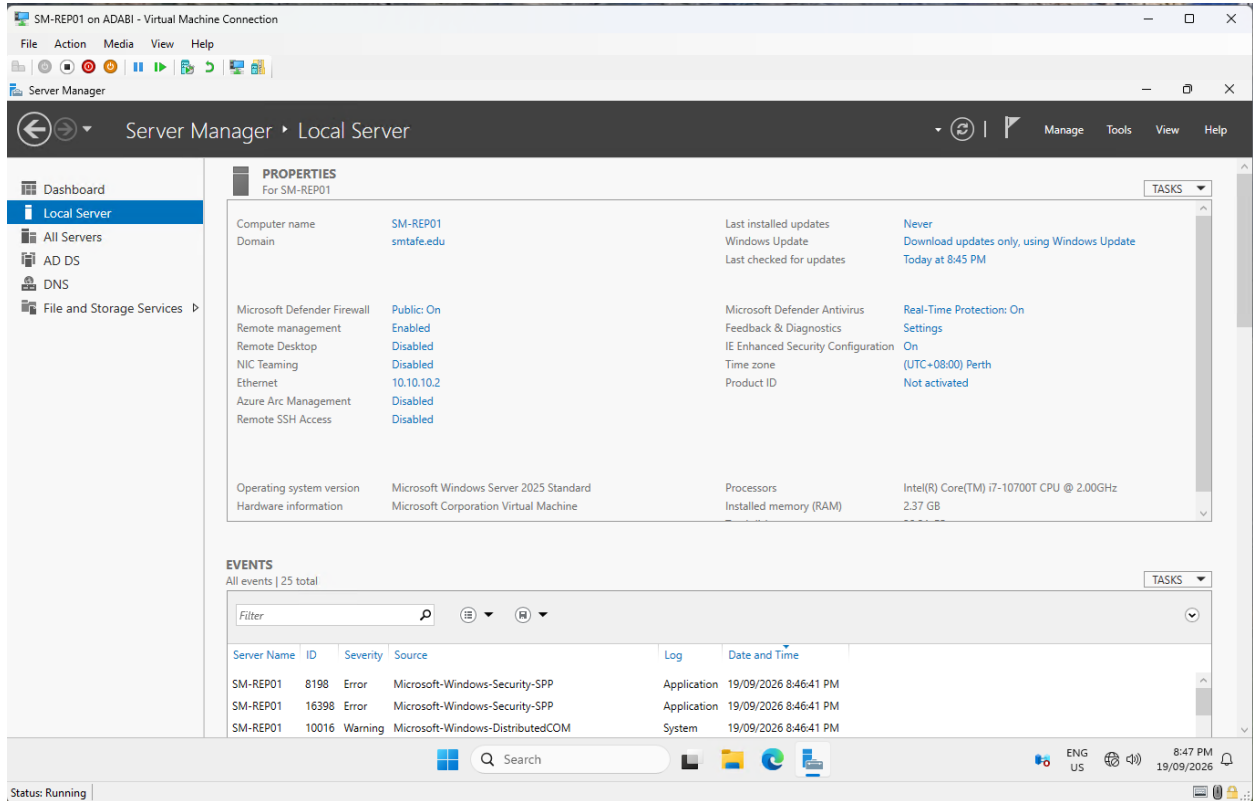


Figure 11-Active Directory Users and Computers

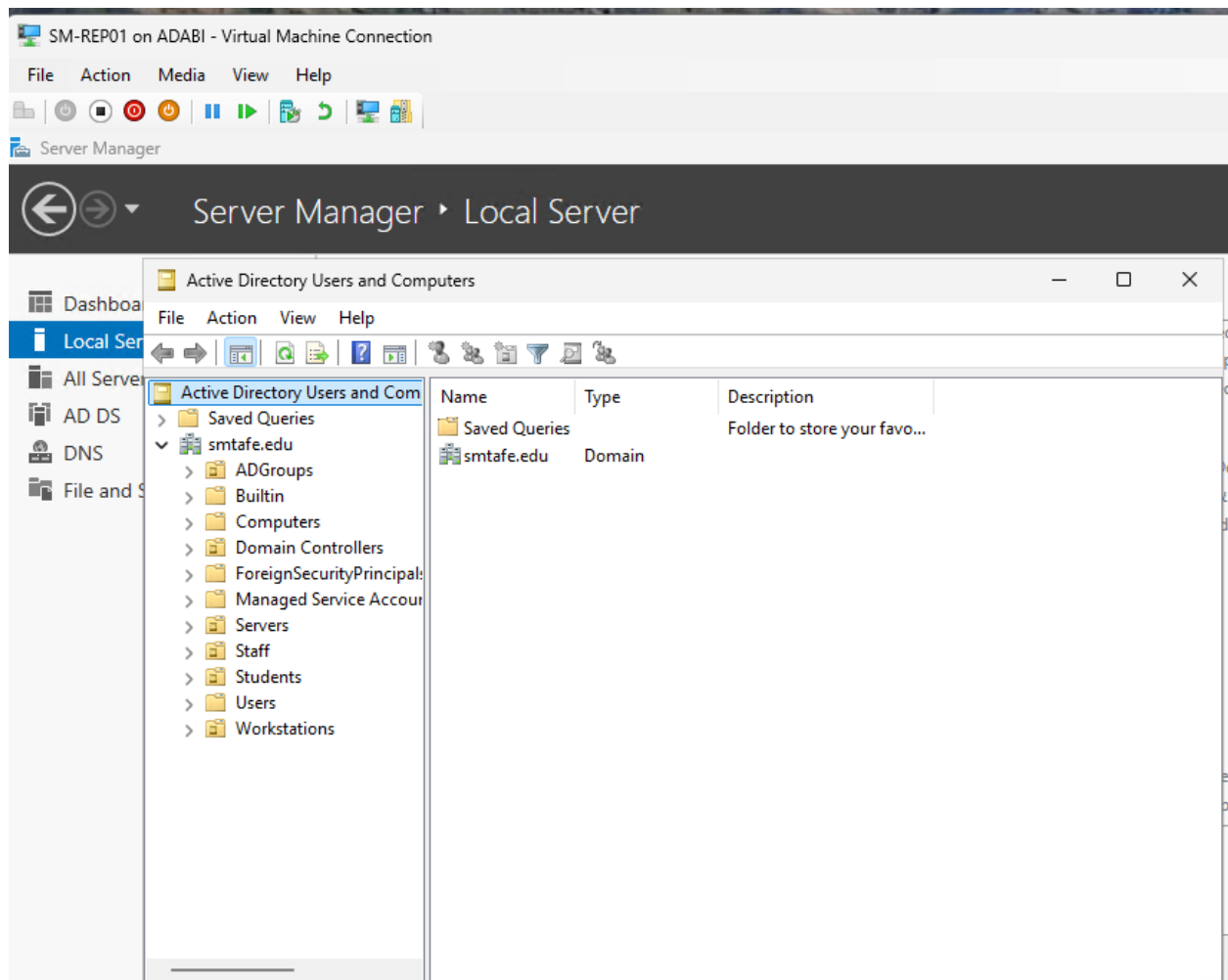


Figure 12-DNS

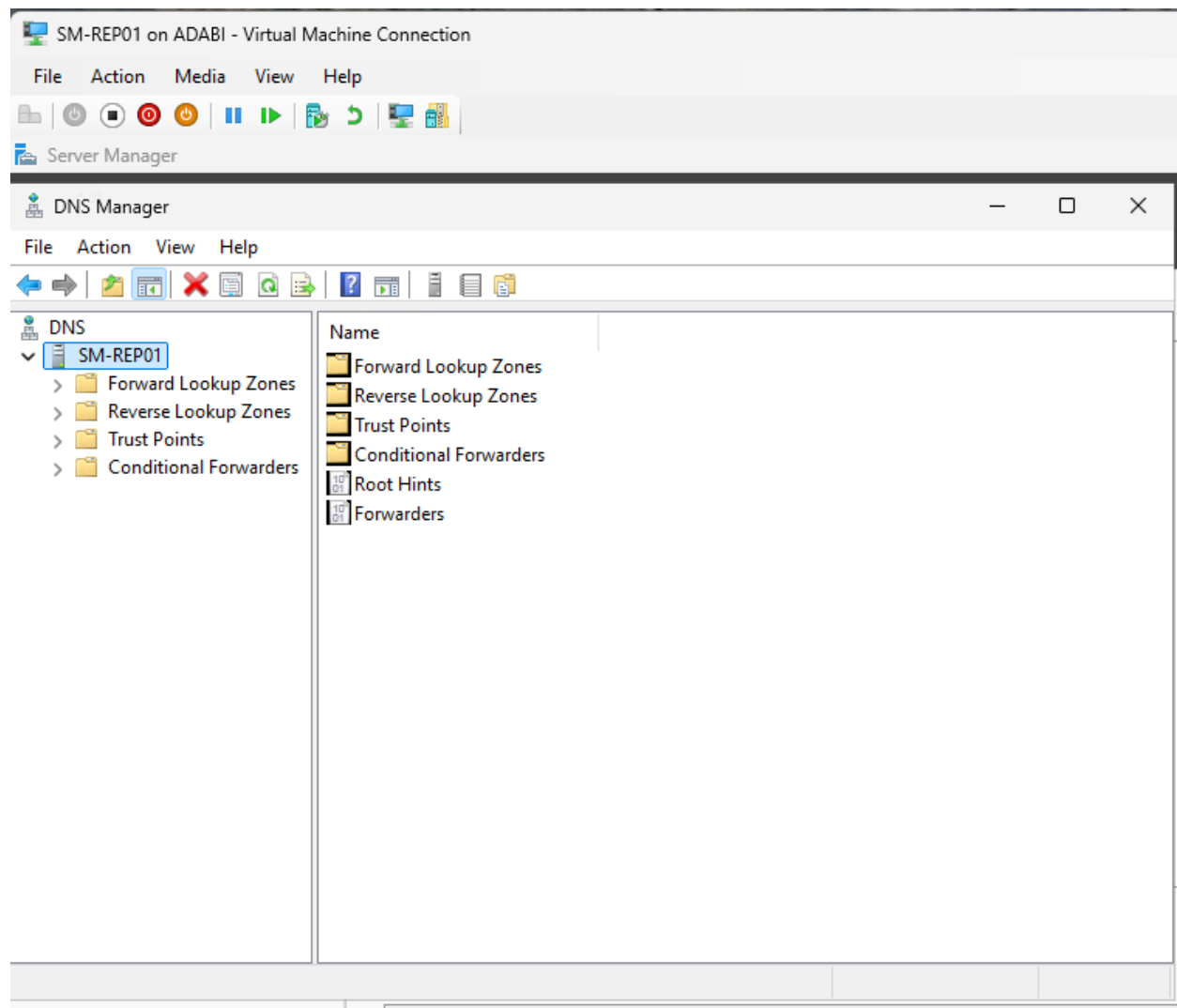


Figure 13-Domain membership

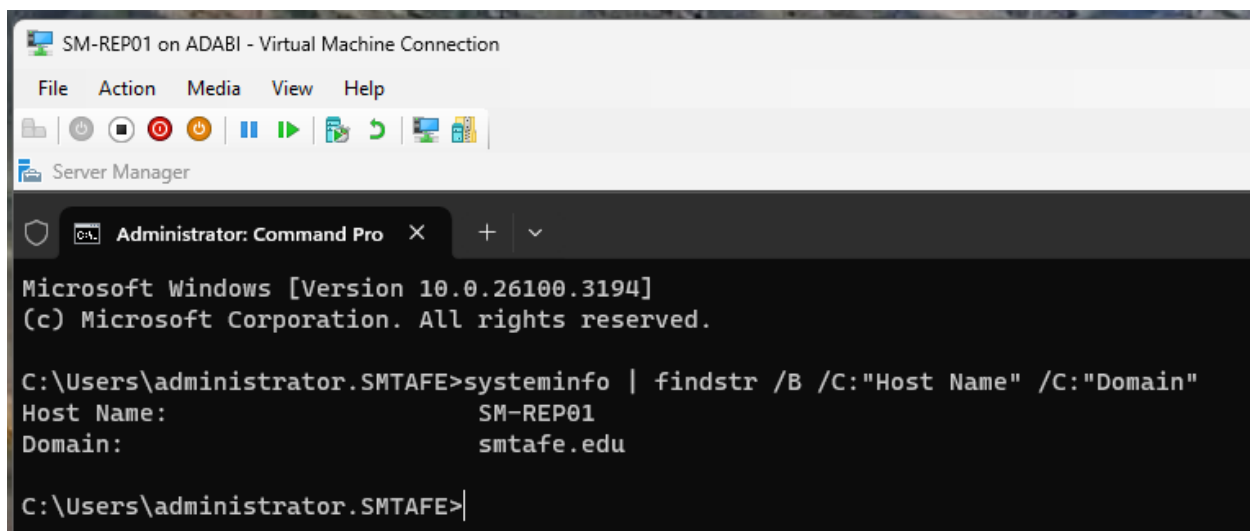
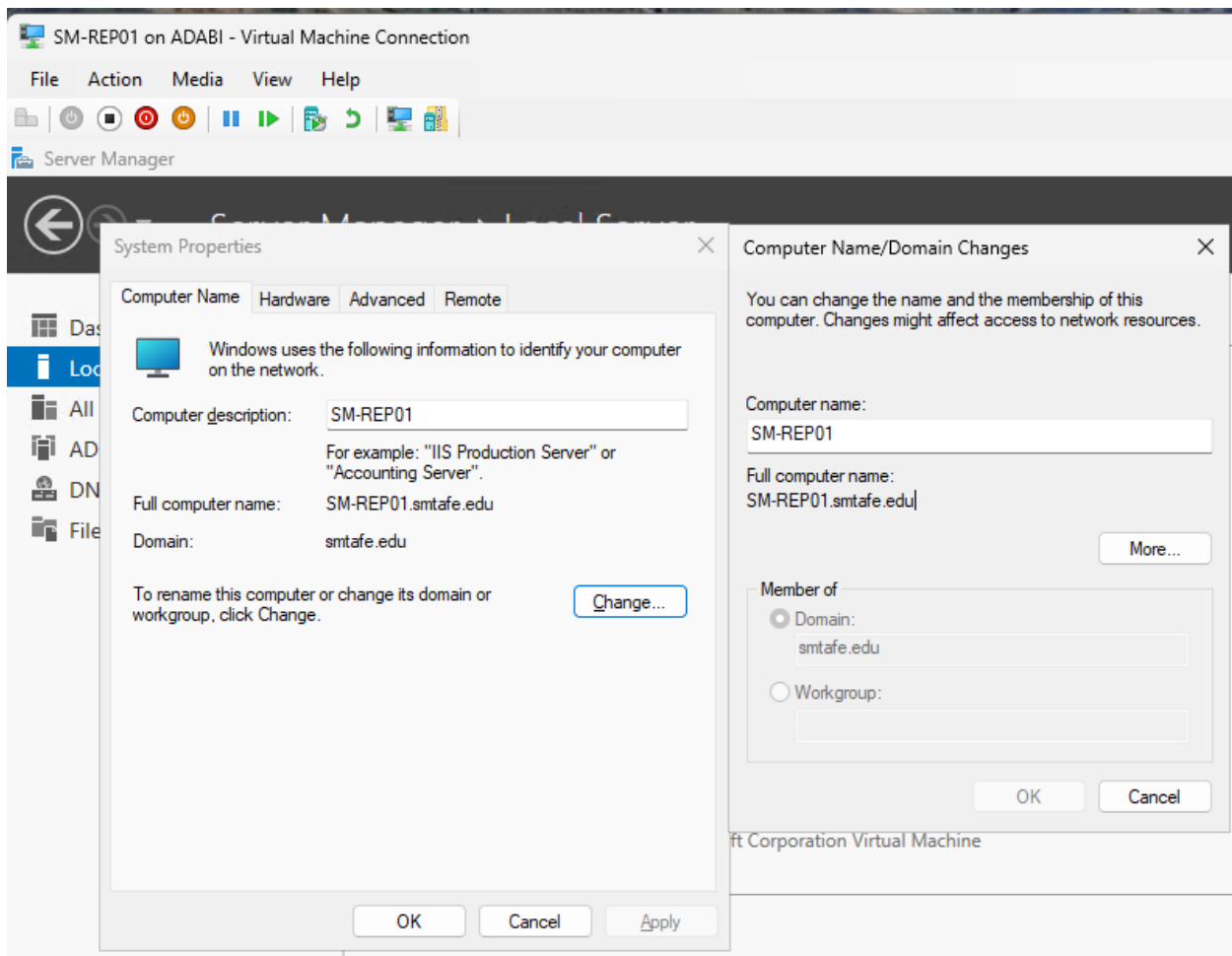


Figure 14-Replication evidence

```
C:\Users\administrator.SMTAFE>repadmin /showrepl

Repadmin: running command /showrepl against full DC localhost
Default-First-Site-Name\SM-REP01
DSA Options: IS_GC
Site Options: (none)
DSA object GUID: 01ea463b-a887-494b-8bde-ba474a5f565c
DSA invocationID: 07419f4e-33e2-46d5-907e-700f7692a74c

==== INBOUND NEIGHBORS =====

DC=smtafe,DC=edu
  Default-First-Site-Name\SM-DC01 via RPC
    DSA object GUID: cd41cb19-c990-4624-a0dd-c4842e3bf370
    Last attempt @ 2026-09-19 21:14:59 was successful.

CN=Configuration,DC=smtafe,DC=edu
  Default-First-Site-Name\NM-DC01 via RPC
    DSA object GUID: 36d69138-477f-49a9-a63d-e652af05e9fe
    Last attempt @ 2026-09-19 20:59:20 failed, result 8524 (0x214c):
      The DSA operation is unable to proceed because of a DNS lookup failure.
    2 consecutive failure(s).
    Last success @ 2026-08-15 20:59:00.
  Default-First-Site-Name\SM-DC01 via RPC
    DSA object GUID: cd41cb19-c990-4624-a0dd-c4842e3bf370
    Last attempt @ 2026-09-19 21:09:53 was successful.
  Default-First-Site-Name\MURD-DC01 via RPC
    DSA object GUID: f3840251-6134-4b0d-ad54-f98c72ce9573
    Last attempt @ 2026-09-19 21:10:05 was successful.

CN=Schema,CN=Configuration,DC=smtafe,DC=edu
  Default-First-Site-Name\SM-DC01 via RPC
    DSA object GUID: cd41cb19-c990-4624-a0dd-c4842e3bf370
    Last attempt @ 2026-09-19 20:59:20 was successful.
  Default-First-Site-Name\MURD-DC01 via RPC
    DSA object GUID: f3840251-6134-4b0d-ad54-f98c72ce9573
    Last attempt @ 2026-09-19 20:59:20 was successful.
  Default-First-Site-Name\NM-DC01 via RPC
    DSA object GUID: 36d69138-477f-49a9-a63d-e652af05e9fe
    Last attempt @ 2026-09-19 21:00:05 failed, result 8524 (0x214c):
      The DSA operation is unable to proceed because of a DNS lookup failure.
```

Replication Evidence – SM-REP01

Active Directory replication on the replica server SM-REP01 was verified using the `repadmin /showrepl` command. The results show successful inbound replication from SM-DC01 for the `smtafe.edu` domain naming context, as well as successful replication of the Configuration and Schema naming contexts. Successful replication from MURD-DC01 was also demonstrated for the relevant naming contexts.

The diagnostic also identified separate replication errors involving NM-DC01, primarily associated with DNS lookup and remote-system/RPC communication. Therefore, the evidence confirms successful

replication paths between SM-REP01 and the available domain controllers, while also documenting an outstanding issue involving NM-DC01.

8. Separate Active Directory Tree

NM-DC01

Figure 15-Server Manager

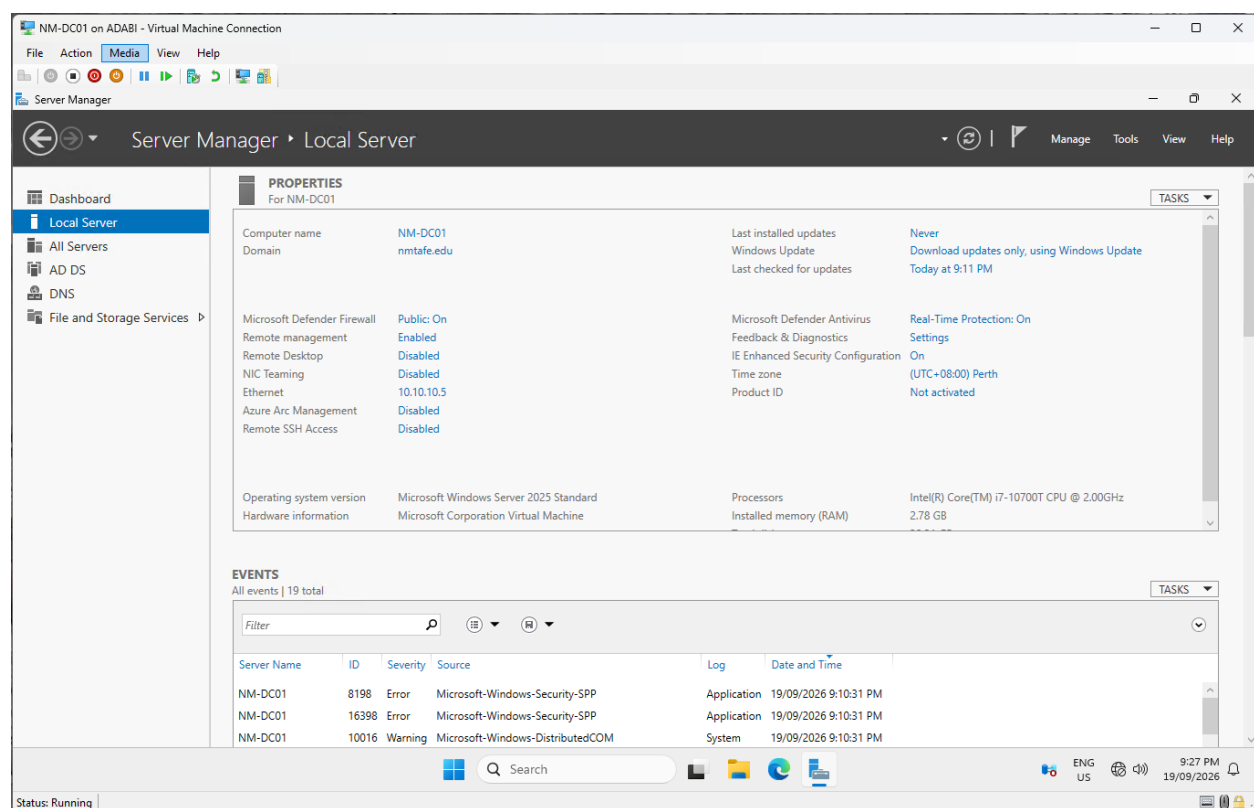


Figure 16-AD Users and Computers

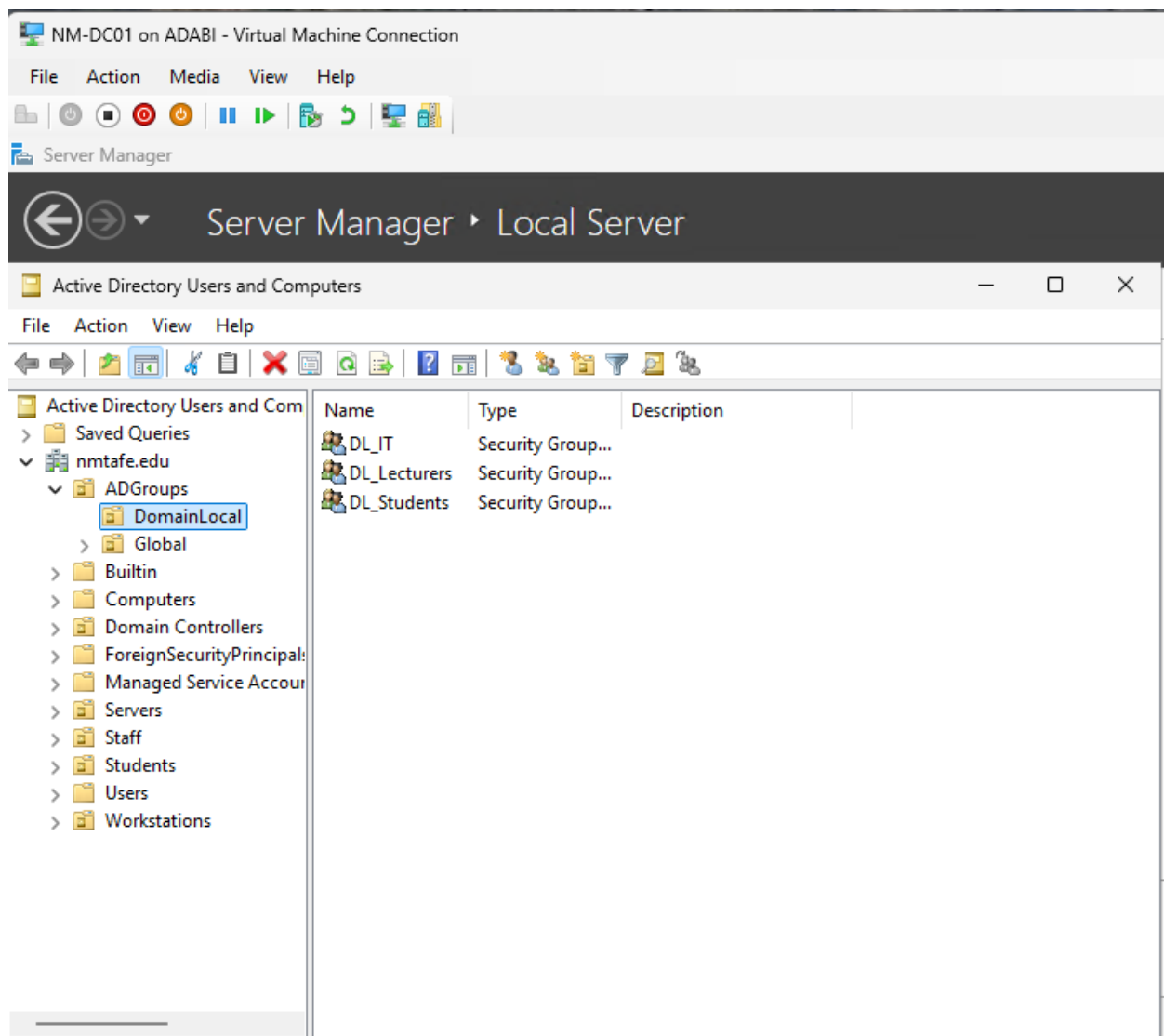


Figure 18-DNS Manager

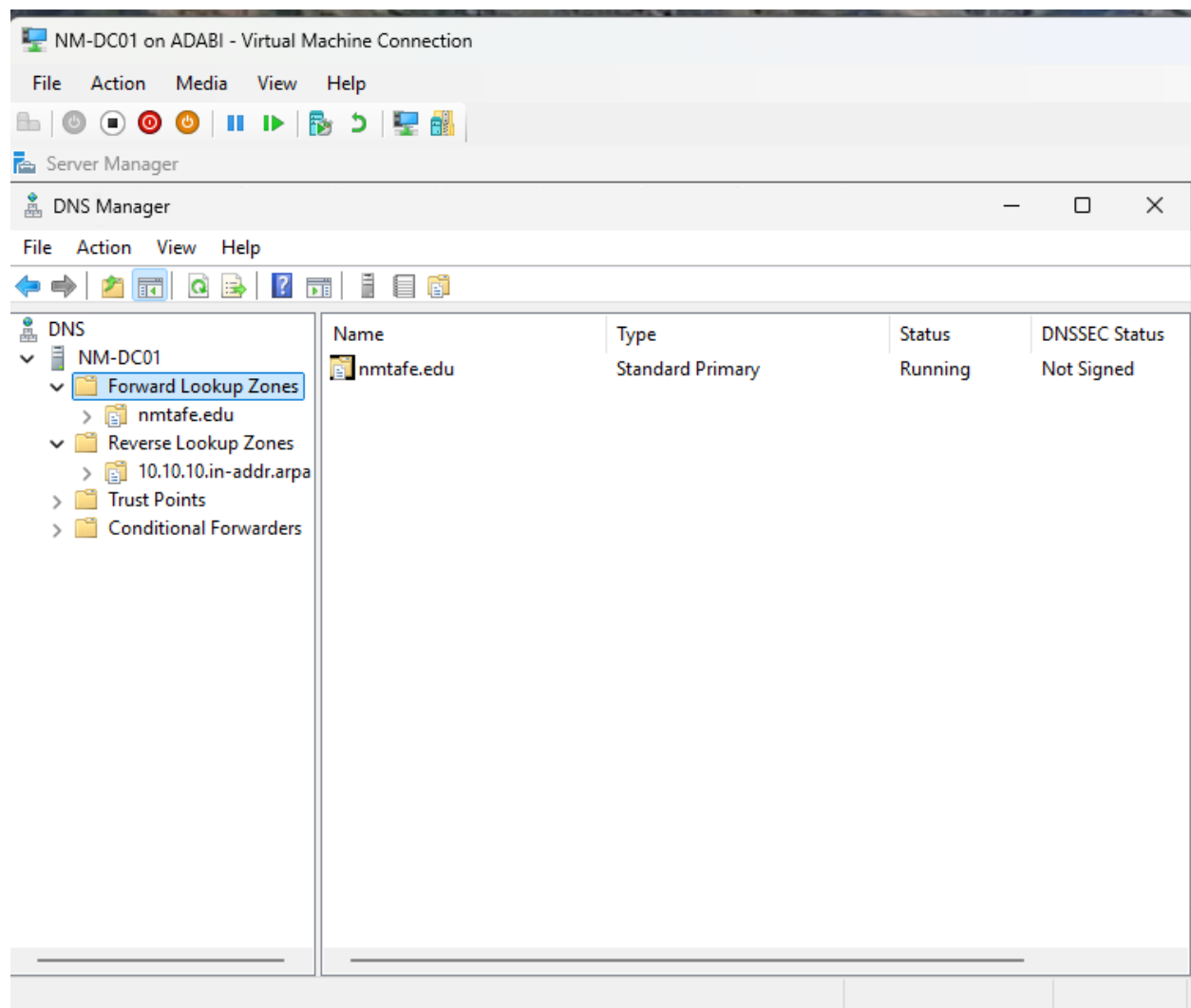


Figure 19-Domain information

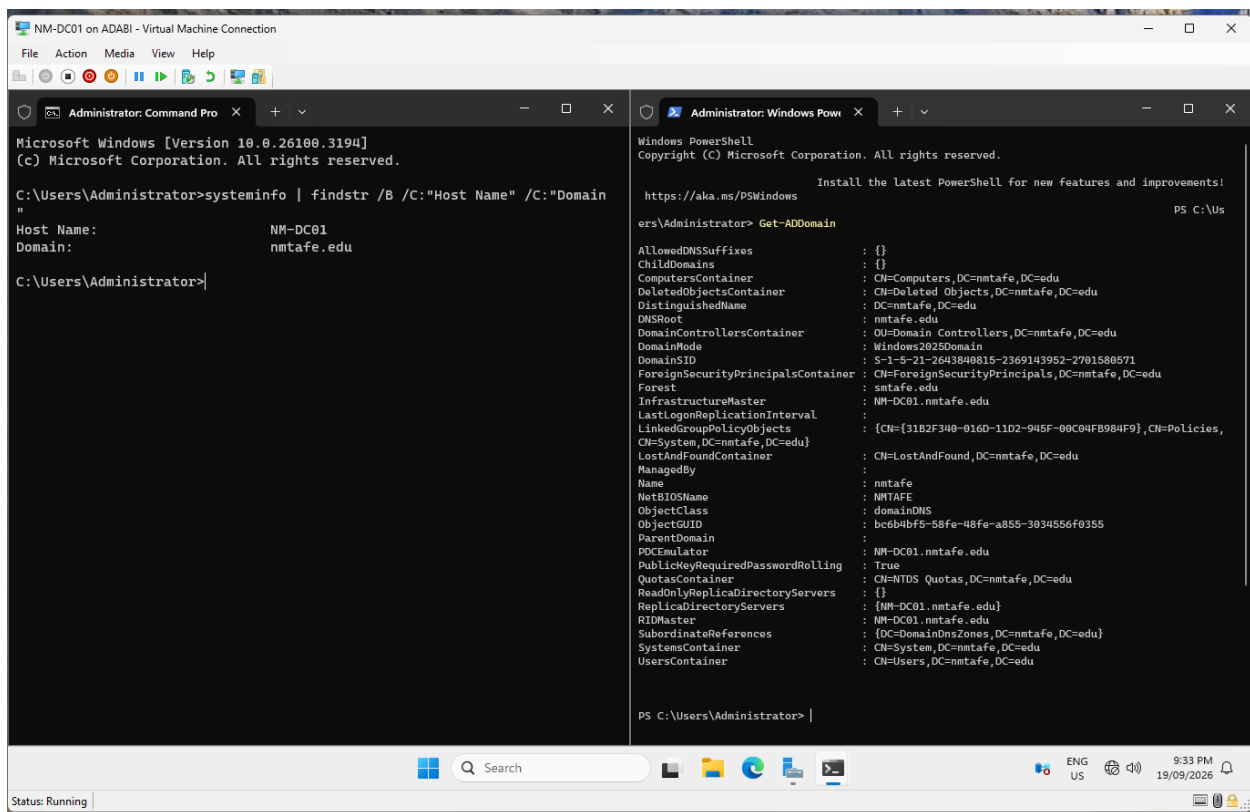
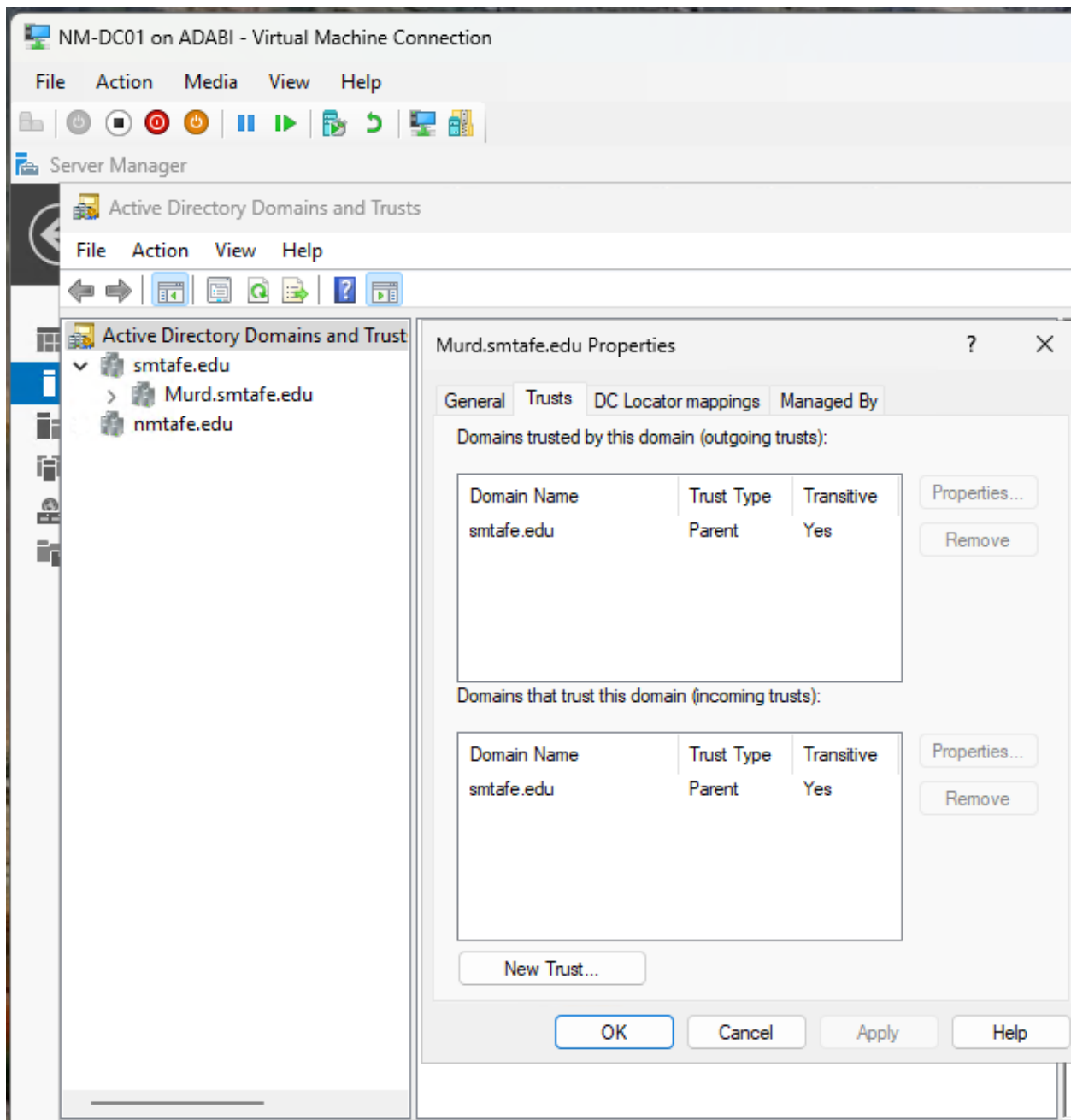


Figure 20-Active Directory Domain and Trust



10. Windows Client Domain Integration

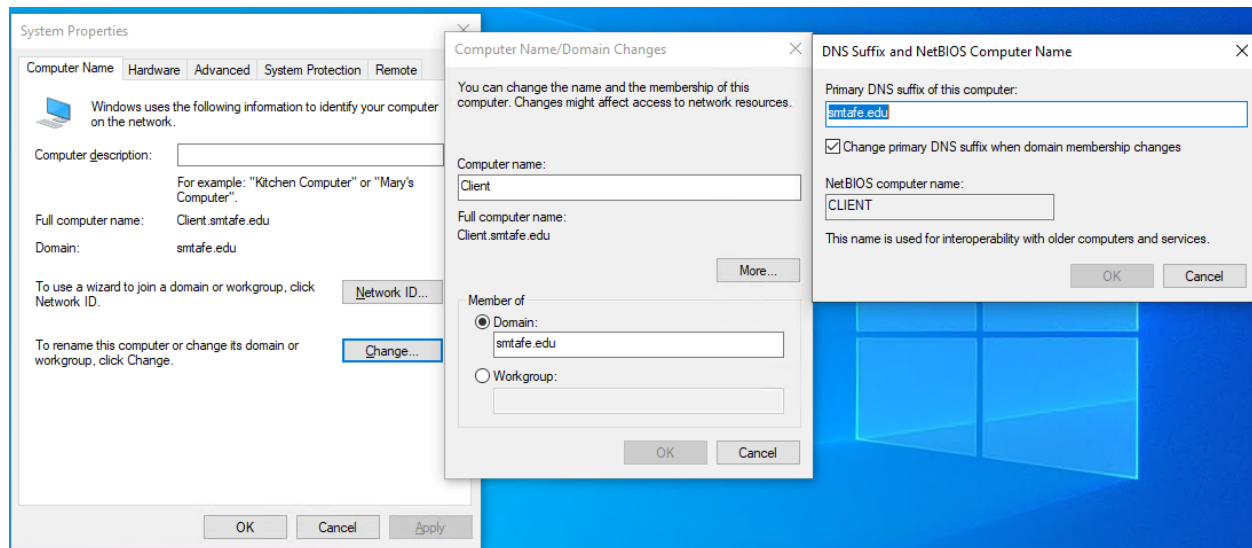
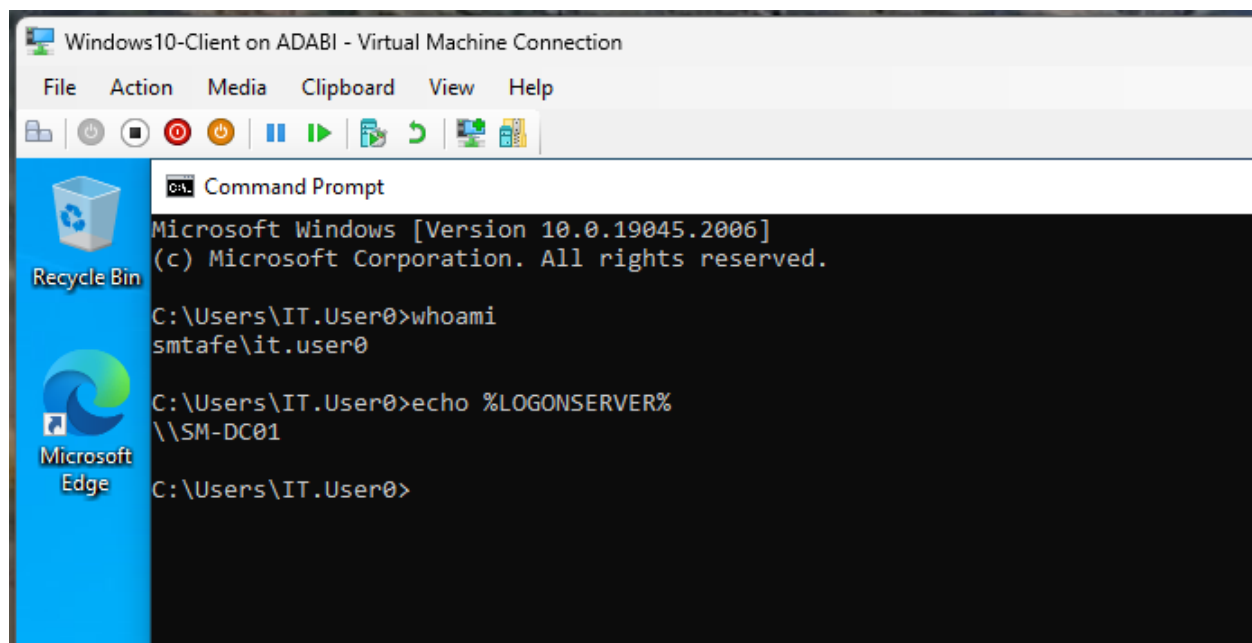


Figure 21-Login screen showing domain authentication.



```

Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.

C:\Users\IT.User0>nltest /dsgetdc:smtafe.edu
        DC: \\SM-DC01.smtafe.edu
        Address: \\10.10.10.10
        Dom Guid: 59dcff62-659c-45b8-baee-41d03f615d63
        Dom Name: smtafe.edu
        Forest Name: smtafe.edu
        Dc Site Name: Default-First-Site-Name
        Our Site Name: Default-First-Site-Name
        Flags: PDC GC DS LDAP KDC TIMESERV WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE FULL_SECRET WS_DS_8 DS_9 DS_
10 KEYLIST 0x40000
The command completed successfully

C:\Users\IT.User0>whoami
smtafe\it.user0

C:\Users\IT.User0>nslookup smtafe.edu
Server:  SM-DC01.smtafe.edu
Address:  10.10.10.10

Name:     smtafe.edu
Addresses: 10.10.10.10
          10.10.10.2

C:\Users\IT.User0>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\IT.User0>

```

Technology	Practical Experience
Microsoft Hyper-V	Virtual machine deployment and management
Windows Server	Server installation and administration
Active Directory DS	Domain controller and directory management
DNS	Forward/reverse DNS configuration and troubleshooting
AD Domains	Root, child and separate tree/domain configuration
AD Users & Groups	User, group and organizational management
Domain Authentication	Windows domain logon and authentication
Windows Clients	Domain joining and administration
Virtual Networking	Hyper-V virtual network configuration
Troubleshooting	DNS, connectivity and domain-service troubleshooting

13. Project Outcome

The completed laboratory demonstrates practical experience in designing and administering a multi-domain Windows Server environment using Microsoft Hyper-V. The environment included a root Active Directory domain, child domain, separate Active Directory tree/domain, replica/secondary server and domain-joined Windows clients. The laboratory provided hands-on experience with Active Directory Domain Services, DNS, domain authentication, user and group administration, Windows client integration, virtualization and infrastructure troubleshooting.